Identification of Synchronicity in Deterministic Chaotic Attractors for the Downscaling Process in the Bogotá River Basin, Colombia

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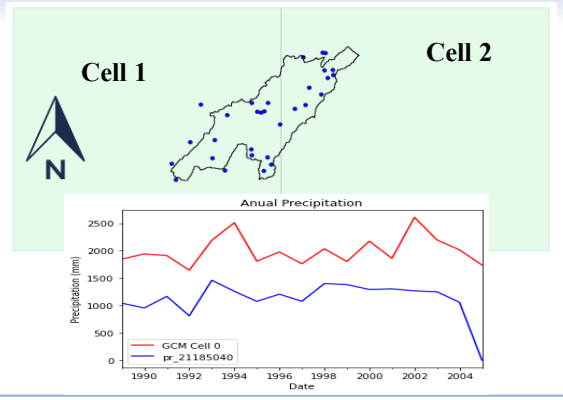
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Due to the complexity of the dynamical climatic system in tropical regions and considering that climatic processes are essentially non-linear and possibly chaotic, predict and use the information of GCMs at the basin scale it's a challenge. This study, reconstruct the phase space of time series at local and regional scale for multiple time steps, identify the presence of deterministic chaos and search for synchronicity between local stations and GCM information, as well as the synchronicity relation between stations.

1. Case of Study

Cell GCM: MPI-ESM-MR (Historical)
N° of cells=2 (1.875°x1.875°)
Period of time: 1850 -2005 (daily)
Mean: 5.13 mm
Standard deviation: 6.12 mm
Maximum: 45.82 mm

Rainfall local stations IDEAM
N° of stations=30
Period of time: 1989 -2019 (daily)
Mean= 2.36 mm
Standard deviation =6.39 mm
Máximum =241.20 mm

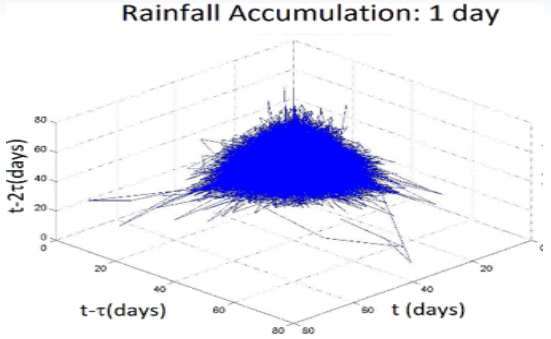


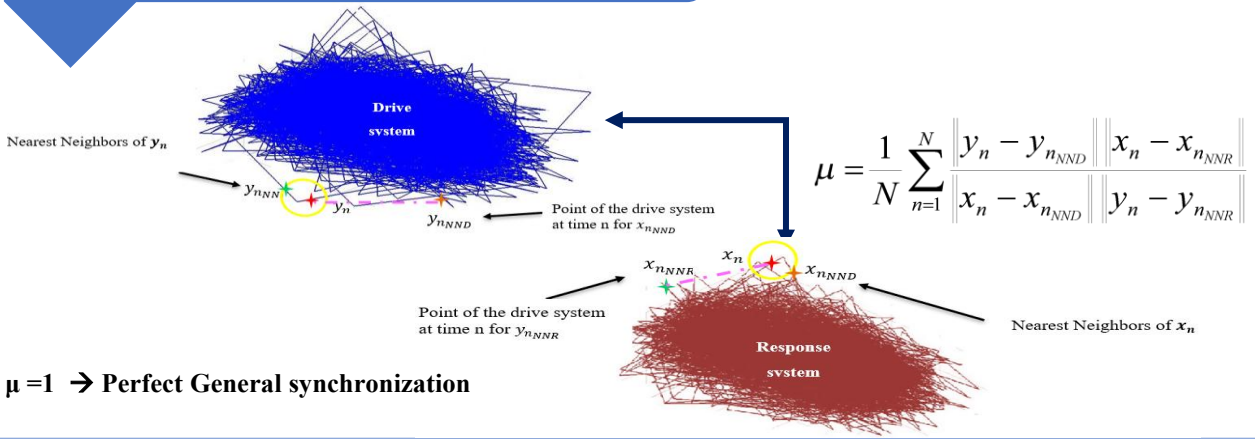
2. Methodology

Phase Space Reconstruction
Method of Time-Delay (Takens , 1981)
Time Delay (τ): - Autocorrelation function
- Mutual Information
Embedding Dimention (*m*): -False Nearest Neighbors

Deterministic Chaos identification
Calculate Lyapunov Spectrum (λ)
(Eckmann et al, 1992)

Synchronicity Search
Compute Mutual False Nearest Neighbors (μ)
(Rulkov, 1995)

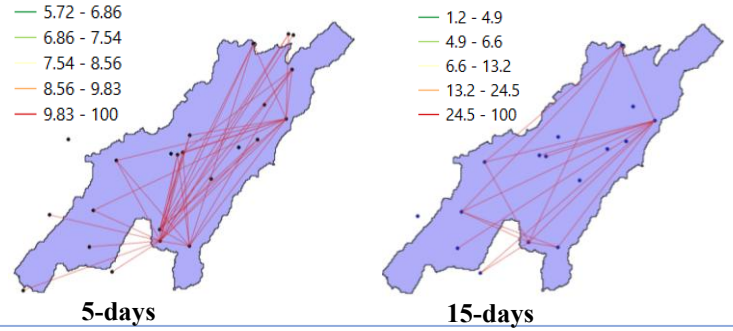




μ =1 → Perfect General synchronization

3. Results

Station's network synchronicity for 5 and 15 step days accumulation



GCM - Station synchronicity

step	Cell 1	Cell 2
5	5.12	5.13
7	4.06	7.22
15	3.41	3.25
30	1.47	1.51

4. Conclusion

These results provide new knowledge about patterns and relationships between climatic variables at different temporal and spatial scales, as well as the potential of synchronicity in chaotic attractors to improve rainfall and temperature downscaling results at aggregated daily-time steps.

References:(1)Eckmann, J.-P., & Ruelle, D. (1992). Fundamental limitations for estimating dimensions and Lyapunov exponents in dynamical systems. Physica D: Nonlinear Phenomena, 56(2), 185–187. https://doi.org/10.1016/0167-2789(92)90023-G. (2) Rulkov, N. F., Sushchik, M. M., Tsimring, L. S., & Abarbanel, H. D. I. (1995). Generalized synchronization of chaos in directionally coupled chaotic systems. Physical Review E, 51(2), 980–994. https://doi.org/10.1103/PhysRevE.51.980 (3)Takens. (1981). Detecting strange attractors in turbulence | SpringerLink. https://link.springer.com/chapter/10.1007%2FBFb0091924

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